

SARASWAT ACADEMY

TEST (CLASS-IX- MATHEMATICS)

CHAPTER- 1: Orienting Yourself: The Use of Coordinates

MCQ'S

Note: Each question has 1 mark

- Q1: The distance of point $(-3, 5)$ from x- axis is?
(a) 4 units (b) 5 units (c) 3 units (d) 8 units
- Q2: In which quadrant the point $(-1, -8)$ lies?
(a) I Quadrant (b) II Quadrant
(c) III Quadrant (d) IV Quadrant
- Q3: The y co-ordinate of a point is distance of that point from :
(a) x-axis (b) y-axis (c) Origin (d) None of these
- Q4: Find the distance between $(-1, 4)$ and $(2, -3)$ is:
(a) 58 units (b) $\sqrt{58}$ units (c) $\sqrt{51}$ units (d) None of these
- Q5: The co-ordinates of one end point of a diameter of a circle are $(4, -1)$ and coordinates of the centre of the circle are $(1, -3)$ then coordinates of the other end of the diameter are :
(a) $(2,5)$ (b) $(-2,-5)$ (c) $(3,2)$ (d) $(-3,-2)$

COMPETENCY BASED QUESTIONS

Note: Each question has 3 marks

- Q1: Write the quadrants for the following points.
(i) $A(3,4)$ (ii) $B(-2,3)$ (iii) $C(-5,-2)$ (iv) $D(4,-3)$
- Q2: Find the distance between:
(i) $(5,3)$ and $(3,2)$ (ii) (a, b) and $(-b, a)$
- Q3: Find the value of x, if the distance between the points $(x, -1)$ and $(3,2)$ is 5.

CASE STUDY

The students of **Green Valley School** were designing a square-shaped flower garden in their playground. Their mathematics teacher asked them to verify whether the four corner points actually form a square before construction.

The four corner points were marked on the coordinate plane as:

A (2, 1), B (-1, 2), C (-2, -1), D (1, -2)

The students plotted these points on the coordinate plane and used the **Distance Formula** and their knowledge of geometry to determine the shape formed.

They also wanted to calculate the total area of the garden.

Q1. Plot the points A, B, C, and D on the coordinate plane.

Q2. Find the length of side AB using the distance formula.

Q3. Find the lengths of BC, CD, and DA.

Q4. Find the lengths of diagonals AC and BD.

Q5. Is ABCD a square? Give reasons.

Q6. Find the area of the square.

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